20269 – Economics of European Integration
Take Home Assignment

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March 2026

1 Part I

Problem I (a)

In 2007, the four tables point to a clear pattern of firm heterogeneity across both countries and sectors. In nearly all cases, means are substantially higher than medians, suggesting right-skewed distributions: which points to most firms being relatively small, while a limited number of very large firms pull the average upward. This suggests that sectoral differences should be interpreted not only through mean values, but also through medians and percentiles.

For France, NACE 22 (*publishing and printing*) includes far more firms than NACE 36 (*furniture*), with 148 firms against only 16. However, the furniture sector appears to operate on a larger scale: mean real revenues, real capital, and real value added are all considerably higher in NACE 36 than in NACE 22. At the same time, the gap in medians is much smaller than the gap in means, which suggests that the French furniture sector is highly concentrated and that a few very large firms drive the average upward. The same pattern broadly emerges for employment, although the number of non-missing observations is limited, especially in French NACE 36. Overall, French furniture appears larger but also much more concentrated than publishing and printing.

The exporter and FDI figures confirm that French NACE 36 is also much more internationally oriented. More than 81% of furniture firms export, compared with only about 35% in publishing and printing. By contrast, FDI activity is very limited in absolute terms in both sectors, so percentage differences should be interpreted with caution, especially given the very small number of firms in French NACE 36.

Table 1: Descriptive statistics for France by sector, 2007

	NACE 22: Publishing and printing					**NACE 36: Furniture**				
Variable	N	Mean	Median	25th Pct.	75th Pct.	N	Mean	Median	25th Pct.	75th Pct.
Real revenues (k€)	141	4706.44	2407.89	1653.51	4430.92	16	13538.84	3193.88	1506.63	11189.29
Real capital (k€)	141	2613.52	1255.82	743.77	2287.57	16	8844.11	1366.91	850.57	7160.68
Employees (L)	83	39.05	24.00	16.00	43.00	9	122.11	45.00	18.00	108.00
Real value added (k€)	139	3138.76	1888.16	1317.98	3615.13	16	7214.29	1980.61	912.76	5562.75

Table 2: Exporter and FDI status for France by sector, 2007

	NACE 22	**NACE 36**
Total firms	148	16
Exporters (N)	52	13
Exporters (%)	35.14	81.25
FDI firms (N)	2	2
FDI firms (%)	1.35	12.50

In Italy, the picture is different. NACE 36 is not only more export-oriented, but also much larger in terms of number of firms, with 209 firms against 104 in NACE 22. Furniture firms display higher real revenues and employment, both on average and at the median, suggesting a broader scale of production. For capital, the median is higher in NACE 36, but the mean is higher in NACE 22, indicating that the publishing and printing sector contains a few very capital-intensive firms. Real value added is not higher in NACE 36; if anything, it is slightly lower than in NACE 22. This implies that the larger scale of Italian furniture firms does not translate into higher value added per firm, possibly because the sector is more input-intensive or operates with lower margins.

Internationalisation is again much stronger in NACE 36. In Italy, 85.17% of furniture firms are exporters, compared with 55.77% in publishing and printing. FDI remains rare in both sectors, which suggests that foreign market integration mainly takes the form of exporting rather than direct multinational activity.

Table 3: Descriptive statistics for Italy by sector, 2007

	NACE 22: Publishing and printing					**NACE 36: Furniture**				
Variable	N	Mean	Median	25th Pct.	75th Pct.	N	Mean	Median	25th Pct.	75th Pct.
Real revenues (k€)	100	5147.80	2034.92	1168.45	3799.10	205	8300.03	2721.90	1507.08	5144.88
Real capital (k€)	100	11145.30	2237.34	1250.84	4746.68	205	8223.07	2838.81	1653.10	5505.23
Employees (L)	92	31.01	16.00	13.00	28.50	192	41.89	22.50	15.00	34.00
Real value added (k€)	100	3983.39	1421.53	870.58	2876.75	205	3811.37	1406.33	825.15	2671.94

Table 4: Exporter and FDI status for Italy by sector, 2007

	NACE 22	**NACE 36**
Total firms	104	209
Exporters (N)	58	178
Exporters (%)	55.77	85.17
FDI firms (N)	2	5
FDI firms (%)	1.92	2.39

Comparing countries, the Italian NACE 22 sector appears more export-oriented than the French one, despite not being uniformly larger across all firm-level indicators. In NACE 36, both countries show very high export participation, but the underlying structures differ: the French sector appears smaller in number and more concentrated, whereas the Italian furniture sector is broader and more diffuse. Overall, the evidence suggests that in 2007 furniture was the more internationally integrated sector in both countries and generally operated on a larger scale, although in France this was driven by a few very large firms, while in Italy the sector was more extensive without showing a clear value-added advantage per firm.

Problem II (a)

Since the sample pools firms from two countries and multiple years, we estimate the production function separately by sector while keeping France and Italy together within each industry, because usually marginal productivities differ across sectors. To improve comparability across observations, we use deflated variables throughout. In the OLS and LP specifications, we include country and year dummies in order to absorb country-specific heterogeneity and common time shocks. In the WRDG specification, we do not add country or year dummies explicitly. Instead, we rely on the Wooldridge proxy-variable estimator, which exploits the panel structure of the data and intermediate inputs as a proxy for unobserved productivity. This allows us to address the simultaneity bias that would affect standard OLS estimates when firms choose inputs after observing their productivity shocks.

Problem II (b)

Table 5 compares the estimated production-function coefficients across OLS, WRDG, and LP for sectors 22 and 36. Overall, the results suggest that OLS tends to deliver larger input coefficients than the proxy-variable estimators, but the evidence of an upward bias in the labour coefficient is not uniform across all methods.

In sector 22, the labour elasticity is equal to 0.415 under OLS, compared with 0.362 under LP. This points to an upward gap of about 0.05 relative to the LP estimate. However, the WRDG labour coefficient is higher, at 0.501. Therefore, for sector 22, the conclusion of an upward OLS bias is clear only when OLS is compared with LP, not when it is compared with WRDG.

In sector 36, the pattern is clearer. The labour coefficient is 0.417 under OLS, 0.363 under WRDG, and 0.320 under LP. Hence, in this sector, OLS gives the highest estimate of the labour elasticity among the three methods.

A similar comparison can be made for the capital coefficient. In sector 22, OLS yields a capital elasticity of 0.555, compared with 0.397 under LP and 0.174 under WRDG. In sector 36, OLS again gives the highest capital coefficient (0.578), followed by LP (0.522) and WRDG (0.473). Overall, OLS tends to attribute a larger share of output variation to observed inputs than the proxy-variable estimators.

The economic reason for a possible OLS bias is simultaneity. Firms observe their productivity shocks before choosing flexible inputs such as labour. If more productive firms choose more labour, labour becomes positively correlated with unobserved productivity in the error term. OLS then attributes part of the productivity effect to labour, which can overestimate its elasticity. Proxy-variable estimators such as LP and WRDG are designed to address this endogeneity problem by using intermediate inputs as proxies for unobserved productivity. For this reason, they are generally preferred to OLS when estimating production functions in the presence of time-varying unobserved productivity, although the magnitude of the correction can differ across estimators and samples.

Table 5: Comparison of production function coefficients for NACE-22 and NACE-36

	NACE-22	**NACE-36**
Lev-Pet		
ln(labour)	0.362***	0.320***
	(0.0581)	(0.0522)
ln(capital)	0.397***	0.522***
	(0.0587)	(0.0959)
WRDG		
ln(labour)	0.501***	0.363***
	(0.0220)	(0.0222)
ln(capital)	0.174***	0.473***
	(0.0653)	(0.0482)
OLS		
ln(labour)	0.415***	0.417***
	(0.0217)	(0.0232)
ln(capital)	0.555***	0.578***
	(0.0159)	(0.0185)
OLS labour coefficient higher than LP	Yes	Yes
OLS labour coefficient higher than WRDG	No	Yes
Observations, OLS and LP	1361	1225
Observations, WRDG	944	870

Standard errors in parentheses.

LP standard errors are bootstrap-based; seeds are fixed for reproducibility.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Problem III (a)

Yes, estimating the production function using revenues rather than value added would generally lead to different results, regardless of whether one uses OLS, Levinsohn–Petrin (LP), or Wooldridge (WRDG). The reason is that the standard Cobb–Douglas production function is a technological relationship defined in physical quantities, whereas firm-level balance-sheet data are typically observed in monetary terms. In its gross-output formulation, the production function can be written as

$$Q_{it} = A_{it} K_{it}^{\beta_K} L_{it}^{\beta_L} M_{it}^{\beta_M},$$

or, in logs,

$$q_{it} = \beta_K k_{it} + \beta_L l_{it} + \beta_M m_{it} + \omega_{it} + \varepsilon_{it},$$

where Q_{it} is physical output, K_{it}, L_{it}, and M_{it} denote capital, labour, and intermediate inputs, ω_{it} is firm-level productivity observed by the firm but not by the econometrician, and ε_{it} is an idiosyncratic shock. This formulation is explicitly about quantities, that is, about how inputs are transformed into output from a technological point of view.

The problem arises because, in practice, output is often proxied by either sales or value added. If revenues are used, then the dependent variable becomes

$$R_{it} = P_{it} Q_{it}, \qquad r_{it} = \ln R_{it} = \ln P_{it} + q_{it}.$$

Thus, a revenue-based specification does not isolate physical production: it combines physical output with firm-specific prices. Any heterogeneity in prices, markups, demand conditions, or product quality is therefore embedded in the dependent variable. As a consequence, the estimated coefficients no longer represent purely technological elasticities; instead, they reflect revenue elasticities. Likewise, the residual captures a form of revenue productivity rather than pure physical productivity. This issue remains relevant even when variables are deflated, because sector-level price indices remove common price movements but do not eliminate firm-level price heterogeneity.

Using value added changes the estimation problem in a different way. Since value added is constructed as revenues net of intermediate input expenditures, it does not coincide with gross output and therefore does not recover the same underlying production function. In particular, once materials are netted out of the dependent variable, the coefficient on intermediate inputs can no longer be estimated and interpreted as a direct production elasticity in the same way as in a gross-output specification. Put differently, a value-added regression focuses on the contribution of capital and labour to net output, whereas a gross-output regression also models the role of intermediate inputs directly.

Moreover, the value-added approach is theoretically valid only under stronger assumptions. To express value added as a function of capital, labour, and productivity alone, one needs a separability assumption: materials must be separable from the other inputs in the production process. If the true production function takes the form

$$Y = F(K, L, M, \omega),$$

then value added can be used cleanly only if output can be decomposed, for example, as

$$Y = G(K, L, \omega) \cdot H(M) \qquad \text{or} \qquad Y = G(K, L, \omega) + H(M).$$

If this condition does not hold, for instance because materials interact with labour or with productivity shocks, value added is not a simple function of (K, L), and the resulting coefficients may be biased. This is why, from a theoretical standpoint, gross output is generally the more complete specification whenever reliable information on intermediate inputs is available.

These issues affect OLS, LP, and WRDG in the same fundamental way. The distinction between revenues and value added concerns the economic object being estimated, not the particular econometric method. OLS suffers from simultaneity because firms observe productivity shocks and adjust inputs accordingly. LP and WRDG were developed precisely to address this endogeneity by using proxy variables such as materials. However, these methods correct for the correlation between inputs and unobserved productivity; they do not solve the additional problem created by

firm-specific output prices. In other words, LP and WRDG can help identify production elasticities in the presence of simultaneity, but they do not transform a revenue-based equation into a purely physical production function.

A further implication is that a revenue-based specification is more exposed to distortions coming from markups and demand conditions. If two firms produce the same physical quantity with the same technology, but one charges a higher price, the revenue-based specification will attribute higher "output" to that firm even though its physical productivity may be identical. Value added partly mitigates this issue by netting out spending on intermediate inputs, but it remains a monetary measure and therefore still incorporates final-output price heterogeneity.

Overall, the choice between revenues and value added is not innocuous. Revenues correspond to output measured in monetary terms and therefore embed prices, markups, and demand conditions. Value added removes the direct contribution of materials from the left-hand side, but it requires stronger assumptions and changes the interpretation of the coefficients. Hence, the two specifications are not equivalent and are likely to generate different estimates in OLS, LP, and WRDG.

Problem IV (a)

Table 6: Number of trimmed TFP observations by estimator and sector

	NACE 22	NACE 36	Total
WRDG	26	24	50
LP	26	24	50

For each estimator and sector, TFP values below the 1st percentile and above the 99th percentile were set to missing. Table 6 shows the number of outliers, by estimation procedure and in each sector. Outliers are not physically dropped from the dataset. Therefore, subsequent analyses automatically use the relevant cleaned sample depending on the TFP measure involved, while preserving the original dataset structure for variables not affected by the TFP trimming.

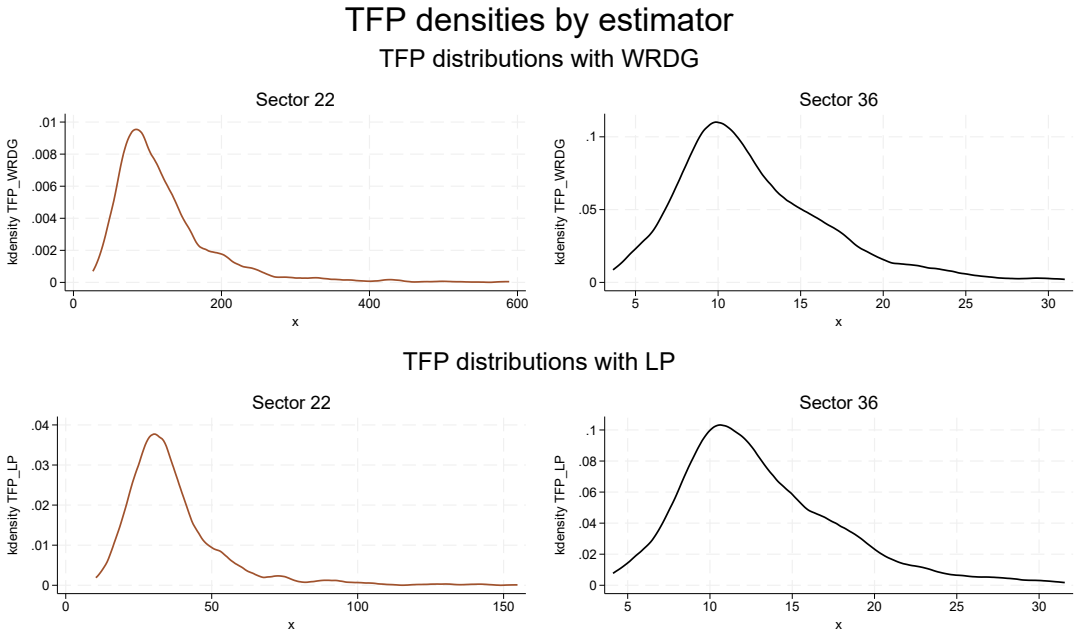


Figure 1: TFP densities comparison by estimator and sector.

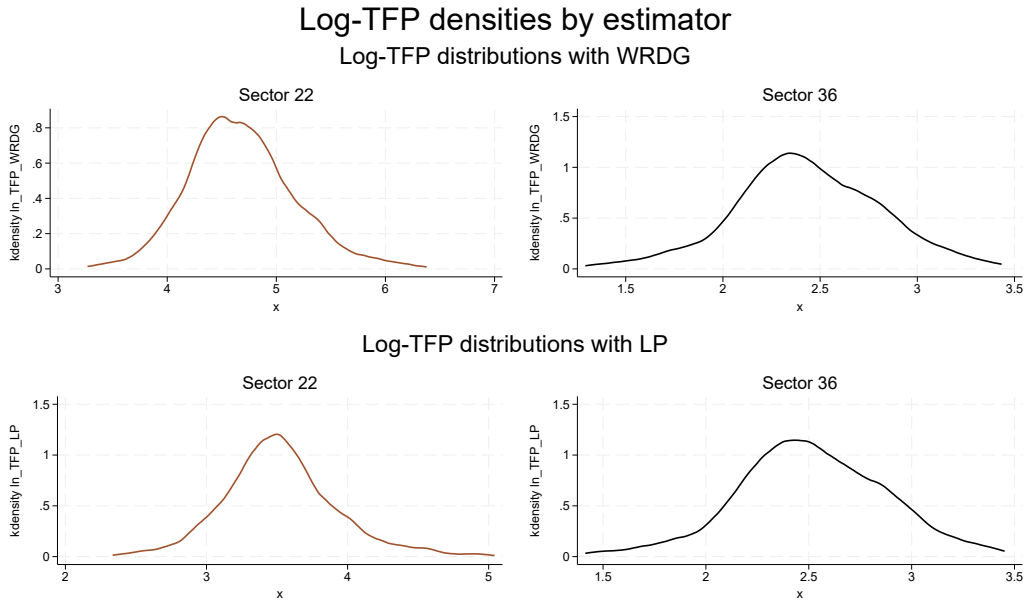


Figure 2: Log-TFP densities by estimator and sector.

Figure 1 provides a visual inspection of the kernel density distributions of firm-level TFP estimates in sectors 22 and 36 under the WRDG and LP procedures. Figure 2 reports the corresponding log-transformed distributions which makes the estimated productivity distributions more regular and easier to compare across sectors. Finally, Figure 3 illustrates the results discussed below by comparing the density distributions across sectors and transformations.

WRDG Under the WRDG estimator, the two sectors display clearly different productivity profiles. In Figure 1, sector 22 exhibits a strongly right-skewed distribution, with most firms concentrated at relatively low TFP levels and a long right tail capturing a limited number of highly productive firms. This suggests substantial heterogeneity within the sector. By contrast, sector 36 is also right-skewed, but its distribution is much more compact and concentrated over a narrower range of values, with a less pronounced upper tail. Hence, as also shown in Figure 3, relative to sector 36, sector 22 appears considerably more dispersed and characterised by a wider spread of productivity outcomes.

The log-transformed distributions in Figure 2 and Figure 3 make this contrast even clearer. Sector 22 remains shifted to the right relative to sector 36, suggesting a higher average productivity level, but it is also flatter and more spread out. Sector 36, instead, shows a more pronounced peak around lower log-TFP values, pointing to a greater concentration of firms around similar productivity levels. Therefore, under WRDG, sector 22 seems to combine higher average productivity with stronger within-sector heterogeneity, while sector 36 appears more concentrated and less dispersed.

LP The LP estimator delivers a very similar qualitative picture. In Figure 1, sector 22 continues to display a markedly right-skewed distribution and a much longer tail than sector 36, whereas sector 36 remains more compact and concentrated. The main difference relative to WRDG is that the LP estimates appear somewhat compressed, especially for sector 22, whose support becomes narrower and whose extreme tail is less pronounced. Nevertheless, the overall sectoral pattern remains unchanged.

This stability is confirmed in Figures 2 and 3. After the log transformation, sector 22 is again located to the right of sector 36 and remains more dispersed, while sector 36 retains a higher degree of concentration around lower log-TFP values. The similarity between WRDG and LP

suggests that the contrast between the two sectors is not driven by the estimation method alone, but reflects a robust difference in the distribution of productivity across industries.

Overall, both estimators point to the same conclusion. Sector 22 is characterised by a wider and more right-skewed productivity distribution, with greater dispersion and a higher average level of log-TFP. Sector 36, by contrast, appears more concentrated around lower productivity values and displays less heterogeneity. While WRDG tends to produce a more pronounced upper tail, especially for sector 22, the substantive interpretation remains unchanged under LP.

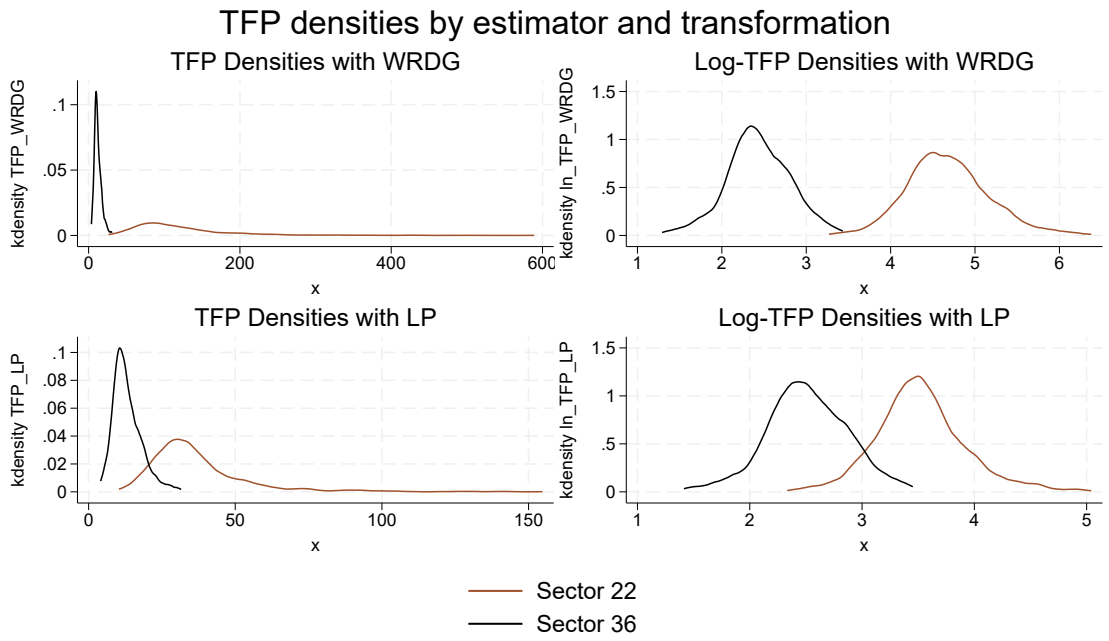


Figure 3: TFP densities comparison by estimator and transformation.

Problem IV (b)

Figure 4 compares the Wooldridge TFP distributions for sector 22 in France and Italy in 2001 and 2007, using the cleaned sample from point (a).

In France, the 2007 distribution is clearly shifted to the right relative to 2001. Both the mean and the median increase, from 110.72 to 142.79 and from 103.41 to 132.08, respectively. At the same time, skewness declines from 1.21 to 1.04, and kurtosis also falls slightly from 4.25 to 3.89. This suggests that the improvement in productivity was relatively broad-based, with the 2007 distribution becoming somewhat less asymmetric.

Italy shows a different pattern. The mean remains broadly stable, increasing only slightly from 117.52 to 118.96, whereas the median declines from 99.27 to 87.07. This indicates that the central part of the distribution shifted left, even though the upper tail remained sufficiently important to keep the mean roughly unchanged. At the same time, skewness decreases from 3.00 to 2.29 and kurtosis from 14.46 to 8.79. Hence, the Italian distribution remains highly right-skewed in both years, but it becomes less extreme in 2007 than in 2001.

Overall, the evidence points to a clear improvement in France, with a rightward shift of the whole distribution and a moderate decline in asymmetry. In Italy, by contrast, changes are more uneven: the centre of the distribution weakens, while the right tail remains relevant, so that average productivity changes little even as the median declines.

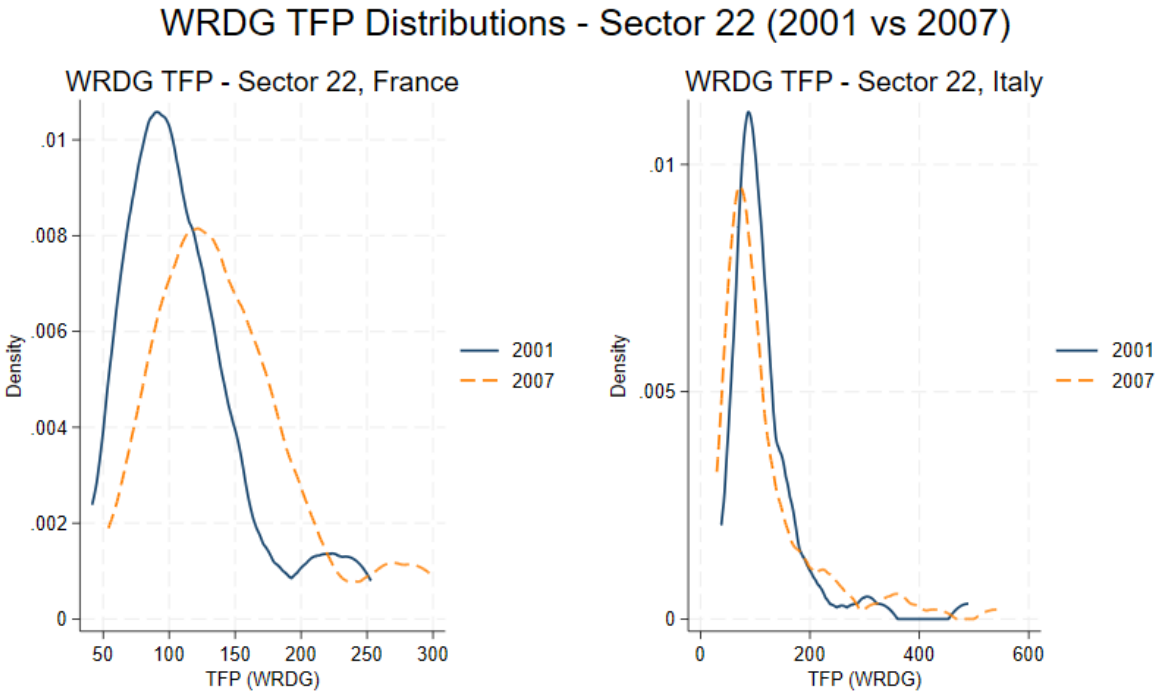


Figure 4: WRDG TFP distributions in sector 22, 2001 vs. 2007.

Table 7: WRDG TFP summary statistics, sector 22

Country	Year	N	Mean	Median	Std. Dev.	Skewness	Kurtosis
France	2001	79	110.72	103.41	46.18	1.21	4.25
France	2007	82	142.79	132.08	55.35	1.04	3.89
Italy	2001	61	117.52	99.27	71.49	3.00	14.46
Italy	2007	89	118.96	87.07	91.62	2.29	8.79

Problem IV (c)

Density plots already suggest that the shifts between 2001 and 2007 are not homogeneous throughout the distribution. The rightward shift is more pronounced in some parts of the distribution than in others, and the change cannot be described as a simple parallel translation of the whole density. Hence, both the location and the shape of the TFP distribution appear to change over time.

A natural parametric model for the upper tail of the TFP distribution is the Pareto distribution, which is widely used in the industrial-organisation and trade literatures. Under this assumption, the survival function satisfies

$$1 - F(\varphi) = \left(\frac{\varphi_{\min}}{\varphi}\right)^{k}, \qquad \varphi \geq \varphi_{\min},\ k > 0. \tag{1}$$

Taking logarithms of both sides yields the linear relationship

$$\ln\big[1 - F(\varphi)\big] = \underbrace{k \ln \varphi_{\min}}_{\alpha} + \underbrace{(-k)}_{\beta} \ln \varphi, \tag{2}$$

where $\beta < 0$. We estimate β by OLS, replacing the theoretical CDF with the empirical within-country $\times$ year CDF constructed from the WRDG TFP estimates.

The estimated Pareto slope coefficient $\hat{\beta}$ is directly interpretable: a less negative $\hat{\beta}$ implies a fatter right tail and therefore greater upper-tail dispersion, whereas a more negative $\hat{\beta}$ implies a thinner tail and less dispersion among the most productive firms.

Tables 8 and 9 report the estimated Pareto slope coefficients for France and Italy in 2001 and 2007.

Table 8: Pareto slope coefficients — France, Sector 22 (WRDG TFP)

Year	$\hat{\beta}$	Std. Error
2001	-2.243	0.101
2007	-2.309	0.111

Table 9: Pareto slope coefficients — Italy, Sector 22 (WRDG TFP)

Year	$\hat{\beta}$	Std. Error
2001	-2.081	0.087
2007	-1.565	0.042

In France, the estimated slope changes from $\hat{\beta}_{2001} = -2.243$ to $\hat{\beta}_{2007} = -2.309$. Equivalently, the implied Pareto shape parameter rises slightly from $\hat{k}_{2001} = 2.243$ to $\hat{k}_{2007} = 2.309$. This suggests a slightly thinner right tail in 2007, and hence marginally lower upper-tail dispersion.

In Italy, the estimated slope changes from $\hat{\beta}_{2001} = -2.081$ to $\hat{\beta}_{2007} = -1.565$. Equivalently, the implied Pareto shape parameter declines from $\hat{k}_{2001} = 2.081$ to $\hat{k}_{2007} = 1.565$. This implies a fatter right tail in 2007, and therefore greater dispersion among the most productive firms.

Overall, the Pareto evidence confirms that the change in the TFP distribution is not homogeneous. France displays at most a modest thinning of the upper tail, whereas Italy shows a clear thickening of the upper tail, consistent with a more uneven and polarized productivity distribution in 2007.

Problem IV (d)

To estimate the export and FDI premia, we first combine the LP TFP estimates for sectors 22 and 36 into a single variable `ln_TFP_LP`, already constructed in Problem IV(a). Outliers in LP TFP were set to missing, rather than physically dropped from the dataset. Therefore, Stata automatically excludes these observations whenever `ln_TFP_LP` is used in graphs, summary statistics, and regressions.

We then define a three-way export status variable:

- **Status 0** — domestic firms only ($exporter = 0$, $FDI = 0$)
- **Status 1** — pure exporters ($exporter = 1$, $FDI = 0$)
- **Status 2** — multinationals/exporters with FDI ($exporter = 1$, $FDI = 1$)

A small number of firms have $FDI = 1$ but $exporter = 0$. These firms do not fit the three-way hierarchy used here and are left with missing export status.

Figure 5 plots the unconditional kernel density of log LP TFP by export status for the combined sample of sectors 22 and 36. At first glance, the graph appears to contradict the productivity pecking order predicted by Melitz (2003), whereby domestic firms should be the least productive, followed by exporters and then multinationals. Instead, domestic firms appear more concentrated at relatively high productivity levels, while exporters and multinationals are shifted to the left.

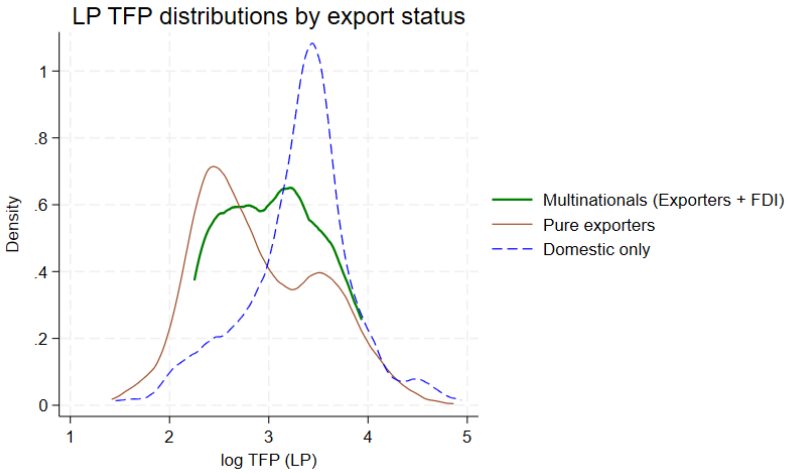


Figure 5: LP TFP distributions by export status (sectors 22 and 36 combined). The apparent reversal of the Melitz pecking order is driven mainly by compositional imbalances across groups.

Diagnostic of compositional bias. This apparent contradiction is mainly driven by compositional imbalance rather than by a genuine reversal of the productivity ordering. The key issue is that the two sectors differ substantially in average productivity. As shown in Table 10, firms in sector 22 have a mean log LP TFP of 3.520, compared with 2.513 in sector 36, a difference of about 1.0 log point. This implies that firms in sector 22 are on average about $e^{1.01} \approx 2.7$ times more productive. This large sectoral productivity gap is the main source of the compositional bias.[1]

Table 10: Mean log LP TFP by sector

Sector	Mean	Median	Std. Dev.	N
22 (Publishing & printing)	3.520	3.500	0.395	1,332
36 (Furniture manufacturing)	2.513	2.496	0.361	1,201
Total	3.043	3.065	0.630	2,533

Restricted to observations with non-missing log LP TFP and export status 0, 1, or 2.

The compositional bias arises because domestic firms and exporters are distributed very differently across sectors. As shown in Tables 11 and 12, 82.6% of domestic firms in the cleaned

[1]Country differences in LP-based TFP are much smaller within sector: France and Italy have mean log-TFP values of 3.491 and 3.572 in sector 22, and 2.565 and 2.509 in sector 36; see Table 19 in the Appendix. Thus, within-sector cross-country gaps are small relative to the cross-sector gap.

LP-TFP sample operate in the more productive sector 22, while 63.8% of pure exporters operate in the less productive sector 36. Since domestic firms are disproportionately concentrated in the high-productivity sector and exporters in the low-productivity sector, the raw density plot mechanically places domestic firms to the right of exporters, even in the absence of any genuine productivity advantage conditional on sector.

Table 11: Composition of the domestic group by country and sector

	Sector 22	**Sector 36**	**Total**
France	495	10	505
Italy	245	146	391
Total	740	156	896

Restricted to domestic firms with non-missing log LP TFP.

Table 12: Composition of the pure exporter group by country and sector

	Sector 22	**Sector 36**	**Total**
France	286	62	348
Italy	283	942	1,225
Total	569	1,004	1,573

Restricted to pure exporters with non-missing log LP TFP.

Export and FDI premia: regression estimates. This is precisely why the graphical comparison is insufficient and why regression analysis is needed. By including year, sector, and country fixed effects, the regressions isolate the conditional export and FDI premia after controlling for compositional differences across time, industries, and countries. Standard errors are clustered at the firm level.

We estimate specifications of the form

$$\ln \widehat{\omega}_{it} = \alpha + \beta D_{it} + \gamma_t + \delta_k + \eta_c + \varepsilon_{it}, \tag{3}$$

where $\ln \widehat{\omega}_{it}$ is log LP TFP, D_{it} is either the exporter dummy or the FDI dummy depending on the specification, γ_t are year fixed effects, δ_k are sector fixed effects, and η_c are country fixed effects.

Tables 13 and 14 report the export and FDI premium estimates.

Table 13 contains the export-premium specifications. Column (1) estimates the productivity premium of all exporters relative to domestic firms. Here, the exporter group includes both pure exporters and exporters with FDI, while the comparison group is domestic firms. Column (2) focuses only on the pure-exporter premium: it compares pure exporters to domestic firms, excluding exporters with FDI from the sample. This column therefore measures whether exporting alone is associated with higher productivity.

Table 14 reports the FDI-premium specifications. Column (1) estimates the FDI or multinational premium relative to domestic firms. It compares exporters with FDI to domestic firms, excluding pure exporters. In this restricted comparison, using FDI or *exporter* as the regressor would give the same coefficient, because status-2 firms have both $exporter = 1$ and $FDI = 1$, while domestic firms have both equal to zero. We use FDI because the coefficient is interpreted as the FDI or multinational premium.

Column (2) compares exporters with FDI directly to pure exporters. Domestic firms are excluded. This specification asks whether, among exporting firms, those that also have FDI are more productive than firms that export only. Therefore, the coefficient measures the additional productivity premium associated with FDI conditional on already being an exporter.

Column (3) compares exporters with FDI to all non-FDI firms, pooling domestic firms and pure exporters in the comparison group. This specification asks whether multinational firms are more productive than firms without FDI, regardless of whether those non-FDI firms export or not.

Turning to the results, the **export premium** in Table 13, column (1), is 0.090 log points and statistically significant at the 1% level. The pure-exporter premium in column (2) is very similar, at 0.087 log points, also significant at the 1% level. These estimates imply that exporters are about 9% more productive than comparable domestic firms, conditional on year, sector, and country fixed effects.

The **FDI premium** relative to domestic firms in Table 14, column (1), is larger, equal to 0.226 log points, implying that exporters with FDI are about 25% more productive than comparable domestic firms. Column (2) shows that exporters with FDI are also more productive than pure exporters: the coefficient is 0.146 log points, implying a productivity advantage of about 16%. Finally, column (3) shows that exporters with FDI are more productive than the broader group of non-FDI firms: the coefficient is 0.181 log points, corresponding to a productivity premium of about 20%.

Overall, the regression evidence restores the expected productivity hierarchy: domestic firms are the least productive, pure exporters are more productive, and exporters with FDI display the highest conditional productivity.

Table 13: Export premia

	(1)	(2)
Variables	**All exporters vs non-exporters**	**Pure exporters vs domestic**
Exporter	0.0903***	0.0865***
	(0.0180)	(0.0183)
Observations	2,533	2,469
R-squared	0.643	0.644
Year FE	YES	YES
Sector FE	YES	YES
Country FE	YES	YES

Robust standard errors in parentheses.
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 14: FDI premia

	(1)	(2)	(3)
Variables	**Multi vs domestic**	**Multi vs pure exporters**	**Multi vs non-FDI firms**
FDI	0.226***	0.146***	0.181***
	(0.0394)	(0.0378)	(0.0365)
Observations	960	1,637	2,533
R-squared	0.520	0.655	0.641
Year FE	YES	YES	YES
Sector FE	YES	YES	YES
Country FE	YES	YES	YES

By Multi we mean Multinational (exporter + FDI). Robust standard errors in parentheses.
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Problem V (a)

We estimate two measures of firm-level markups: the Price-Cost Margin (PCM), an accounting-based measure, and the De Loecker-Warzynski (DLW) markup, which is grounded in production function estimation.

PCM markup. The Price-Cost Margin is defined as:

$$\text{PCM}_{it} = \frac{\text{Sales}_{it} - \text{Variable Costs}_{it}}{\text{Sales}_{it}} \tag{4}$$

where variable costs include materials and the wage bill. The implied markup is then:

$$\hat{\mu}_{it}^{\text{PCM}} = \frac{1}{1 - \text{PCM}_{it}} \tag{5}$$

DLW markup. Following De Loecker and Warzynski (2012), the markup can be recovered from the ratio of the output elasticity of labour — estimated via the LP production function — to the observed labour share of revenue:

$$\hat{\mu}_{it}^{\text{DLW}} = \frac{\hat{\beta}_L}{\alpha_{it}^L}, \quad \text{where} \quad \alpha_{it}^L = \frac{W_{it}}{\text{Sales}_{it}} \tag{6}$$

The output elasticity of labour is recovered from the LP estimates in Problem II(a): $\hat{\beta}_L = 0.362$ for sector 22 and $\hat{\beta}_L = 0.320$ for sector 36.
(Both markup measures are cleaned by trimming observations below the 1st and above the 99th percentile of the respective distribution.)

Results. Figure 6 plots the kernel density of both markup distributions after cleaning.

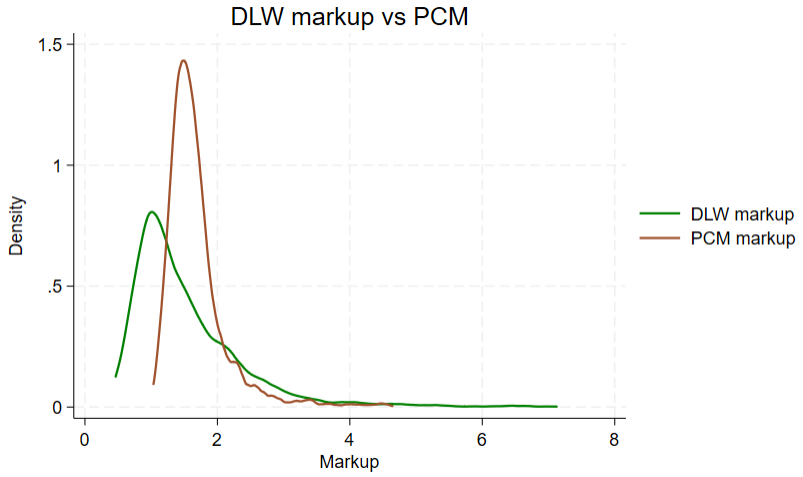


Figure 6: DLW markup vs. PCM markup distributions (sectors 22 and 36).

Is there a bias? Yes. The PCM markup is higher than the DLW markup on average: the mean PCM markup is 1.68 compared to 1.54 for DLW, and the median PCM markup is 1.58 compared to 1.30 for DLW. The PCM distribution is also more concentrated, with an interquartile range of 1.40–1.80, while the DLW distribution is wider (IQR: 0.97–1.86), reflecting greater firm-level heterogeneity in the production-function-based estimates. This difference is consistent with the logic of De Loecker and Warzynski (2012), who recover markups from the firm's cost-minimisation first-order condition rather than from an accounting margin. Accounting-based markups can be distorted by the definition of variable costs: if some capital-related costs vary with capacity utilisation, excluding them from variable costs may understate marginal costs and mechanically overstate the PCM markup. If part of capital costs is related to capacity utilisation, omitting it can understate marginal costs and therefore overstate the markup. The DLW method instead recovers markups from the first-order condition of cost minimisation, using the estimated output elasticity and the observed labour share. Both distributions display right skew, consistent with substantial heterogeneity in market power across firms.

Problem V (b)

We next examine whether exporters charge higher markups than non-exporters, and whether firms with higher markups are also more productive. To do so, we run three regressions. First, we regress the DLW markup on the exporter dummy, controlling for year and sector fixed effects. Second, we add firm size, proxied by log employment, to verify whether the exporter–markup relationship is partly capturing the fact that exporters tend to be larger firms. Third, we regress log Wooldridge productivity on the DLW markup.

The results are reported in Table 15. In column (1), the coefficient on the exporter dummy is positive and statistically significant: exporters have, on average, DLW markups about 0.214 points higher than non-exporters, conditional on year and sector effects. In column (2), once firm size is added, the exporter coefficient decreases slightly to about 0.188 but remains strongly significant. This suggests that part of the exporter markup premium is explained by firm size, but the premium remains substantial even after controlling for size. Firm size itself enters positively, indicating that larger firms tend to charge higher markups.

Column (3) shows a strong positive and statistically significant relationship between DLW markups and Wooldridge productivity. The estimated coefficient is about 0.191, implying that firms with higher markups also tend to display higher measured productivity, conditional on year and sector fixed effects. This pattern is consistent with the idea that more productive firms face lower marginal costs and are therefore able to sustain higher markups. At the same time, the estimate should be interpreted as a conditional correlation rather than as a causal effect of markups on productivity.

Table 15: Exporter status, DLW markups, and WRDG productivity

	(1)	(2)	(3)
Variables	**Markup: no size**	**Markup: + size**	**TFP on markup**
Exporter	0.214***	0.188***	
	(0.0287)	(0.0344)	
log firm size (employment)		0.0429**	
		(0.0217)	
DLW markup			0.1907***
			(0.0115)
Constant	1.381***	1.326***	4.397***
	(0.0505)	(0.0902)	(0.0279)
Observations	3,703	2,735	2,496
R-squared	0.073	0.073	0.889
Year FE	YES	YES	YES
Sector FE	YES	YES	YES

Robust standard errors in parentheses

$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$

2 Part II

Problem VI (a–c)

Figure 7 reports the regional distribution of the average China shock, while Figure 8 reports the pre-sample manufacturing employment share in 1988.

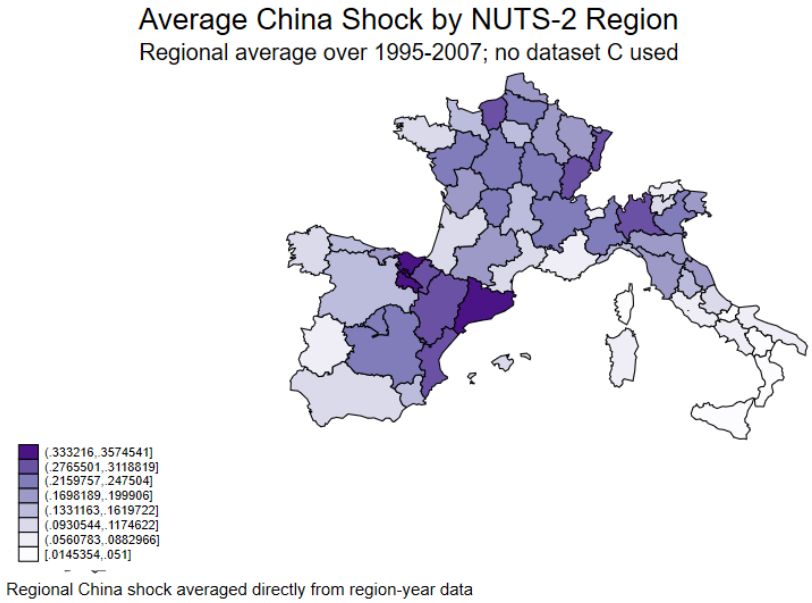


Figure 7: Average China shock by NUTS-2 region. Darker colour reflects higher exposure.

Comparing Spatial Patterns of Shock and Industry

Figure 7 reports the regional distribution of the average China shock, while Figure 8 reports the pre-sample manufacturing employment share in 1988. Taken together, the two maps reveal a broad and economically meaningful spatial overlap. Regions that started with a stronger manufacturing base also tend, on average, to display a higher China shock. This is exactly the type of pattern one would expect from the construction of the shock measure. By design, the regional China shock is built by weighting the growth of Chinese imports at the country-industry level by the initial employment composition of each region. As a consequence, regions with a larger manufacturing workforce at the baseline are, all else equal, more exposed to the subsequent rise in import competition from China. In this sense, the visual correlation between the two maps is not accidental, but reflects the economic logic underlying the Colantone–Stanig measure.

The overlap is especially clear in a number of industrial areas that appear relatively dark in both maps. More manufacturing-intensive regions, particularly in the northern part of Italy and in parts of eastern and northern France, tend also to be among the regions with comparatively stronger China shock exposure. The same is true for some Spanish industrial areas. This broad alignment suggests that the adjustment costs generated by globalization were not spread evenly across space, but instead were concentrated in those regions whose economic structure was more closely tied to manufacturing activities. The maps therefore provide a useful visual counterpart to the theoretical intuition in Colantone and Stanig: trade shocks hit harder where the local economy was more dependent on the sectors most exposed to Chinese import penetration.

At the same time, the two maps do not coincide perfectly, and this is an equally important result. A region's exposure to the China shock depends not only on the size of its manufacturing base, but also on its composition. Two regions can have a similar overall manufacturing share and still be affected very differently if one specializes in sectors such as textiles, clothing, footwear, furniture, or other labour-intensive industries, while the other is more concentrated in sectors less directly challenged by Chinese imports over the sample period. This point is central in the

Colantone–Stanig framework: the shock is stronger not simply where manufacturing is large, but where workers were initially employed in those specific industries for which Chinese import growth subsequently accelerated the most.

For this reason, some deviations between the two maps are entirely expected. Certain regions with a sizeable manufacturing workforce display only moderate China shock intensity, whereas other regions with a less extreme manufacturing share still appear relatively exposed. These discrepancies should not be interpreted as contradictions. Rather, they indicate that within-manufacturing specialization matters substantially. The manufacturing-share map captures the scale of industrial employment, while the China-shock map captures the interaction between scale and sectoral vulnerability. In other words, the first map tells us how industrial a region was at baseline; the second tells us how exposed that particular industrial structure was to the rise of Chinese import competition.

This distinction is important from both an empirical and an economic perspective. Empirically, it explains why the association between the two maps is strong but incomplete. Economically, it highlights that the effects of globalization were not driven by manufacturing presence alone, but by the type of manufacturing in which regions specialized. A region heavily concentrated in industries that were only weakly affected by Chinese competition would not necessarily face a large import shock, even if manufacturing accounted for a large share of local employment. Conversely, a region with a somewhat smaller manufacturing base could still record a substantial China shock if its employment was concentrated in sectors where Chinese import growth was particularly rapid.

Overall, the evidence from the maps is consistent with the theoretical expectations. The broad overlap confirms that the China shock predominantly affected manufacturing regions, which is exactly what one should expect given the way the measure is constructed. At the same time, the visible mismatches confirm that industrial composition matters in addition to industrial size. The two maps therefore convey a coherent message: the geography of the China shock was closely related to the geography of manufacturing employment, but not mechanically so. Exposure to globalization depended not only on whether a region was industrial, but also on what that region produced. This is fully in line with the Colantone-Stanig interpretation of trade competition as a localized and sector-specific source of regional economic pressure.

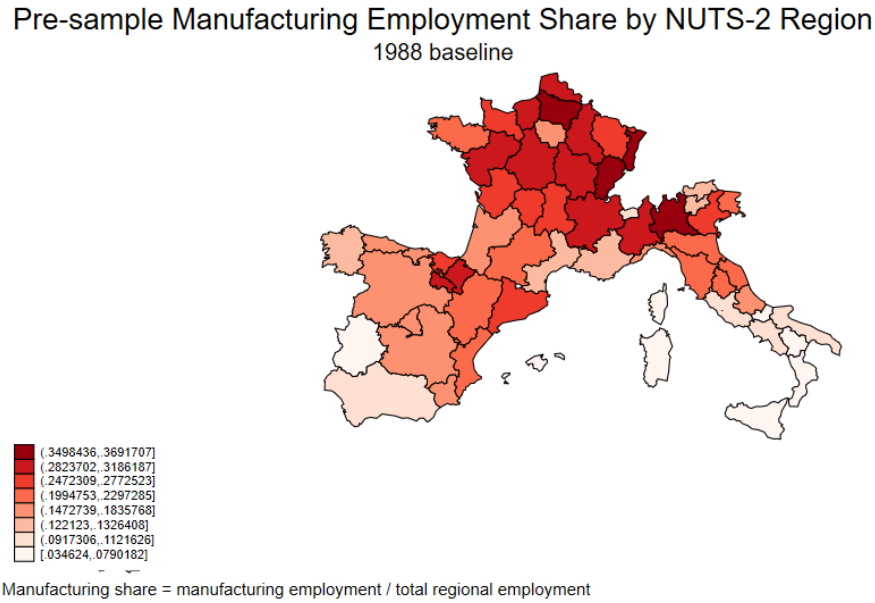


Figure 8: Pre-sample manufacturing employment share by NUTS-2 region (1988 baseline). Darker colour reflects higher share.

Problem VII (a)

Problem VII (a)

We restrict the ESS Round 8 dataset to French respondents and keep the variables required by the assignment: survey weights, gender, age, education, NUTS-2 region of residence, party voted for in the last national election, and the perception variable `sbeqsoc`. We then merge the ESS data with the region-level average China shock constructed in Problem VI.

Since the NUTS-2 codes used in the regional shock dataset correspond to an older French classification, we first recode them into the NUTS-2 codes used in ESS Round 8. This allows us to match all 21 metropolitan French NUTS-2 regions. The merge is successful for all French ESS respondents: 2,070 observations are matched to a regional China shock. After excluding observations with missing values in the regression variables, the final estimation sample used in Problems VII(b)–(c) contains 2,043 respondents.

Problem VII (b)

The weighted OLS regression of `sbeqsoc` on the region-level `china_shock`, controlling for gender, age, education dummies, and NUTS1 regional dummies, and clustering standard errors at the NUTS2 level, points to a statistically significant relationship between trade exposure and attitudes toward the equalizing role of social benefits and services.

In particular, the coefficient on `china_shock` is negative (-0.796) and statistically significant at the 5% level ($p = 0.039$). The corresponding 95% confidence interval, approximately $[-1.546, -0.046]$, excludes zero. This suggests that, conditional on the included controls, individuals living in regions more exposed to the China shock tend to report lower values of `sbeqsoc`.

Since `sbeqsoc` is coded from 1 (agree strongly) to 5 (disagree strongly), a lower value indicates stronger agreement with the statement that social benefits and services lead to a more equal society. Therefore, the negative coefficient implies that greater regional exposure to the China shock is associated with a more favourable view of the equalizing role of welfare. Substantively, this is consistent with the idea that individuals in more trade-exposed and economically vulnerable areas may place greater value on redistribution and social protection.

The model's explanatory power remains limited but non-negligible, with $R^2 \approx 0.056$ and adjusted $R^2 \approx 0.038$. Among the controls, the female dummy is positive and statistically significant, while age is not statistically significant. Some broad regional effects, captured by NUTS1 dummies, are also significant, indicating that territorial heterogeneity remains relevant for welfare attitudes. At the same time, the regression relies on a relatively small number of clusters at the NUTS2 level, so inference should still be interpreted with some caution.

Despite the statistical significance of the coefficient of interest, the OLS estimate should not be interpreted as causal. Several endogeneity issues may still bias the estimated relationship.

Factors that could bias the coefficient downward

1. **Compensatory demand for redistribution in trade-exposed regions:**
   Regions more heavily exposed to Chinese import competition may also be regions where workers experienced greater economic insecurity, job displacement, or wage pressure. In such contexts, individuals may become more supportive of welfare policies and more likely to believe that social benefits help reduce inequality. This would generate a more negative estimated coefficient, even if the China shock itself were not the true causal driver.

2. **Policy feedback from local social protection measures:**
   Areas hit more strongly by industrial decline may also have been more exposed to compensatory public interventions, such as unemployment support, retraining schemes, or broader social assistance. Direct experience with these policies may strengthen the perception that welfare has an equalizing role, again pushing the OLS estimate downward.

Factors that could bias the coefficient upward

1. **Fiscal stress and declining quality of public services in trade-exposed regions:** Regions more heavily affected by import competition may suffer from industrial decline, a shrinking local tax base, and tighter public budgets. Even when welfare spending is present, residents in these areas may perceive social benefits and public services as underfunded, ineffective, or unable to offset growing inequality. If this perception is correlated with exposure to the China shock, the estimated coefficient would be biased upward, because part of the observed effect would reflect dissatisfaction with weak local welfare capacity rather than the causal impact of trade exposure itself.

2. **Selective out-migration from economically distressed regions:** More exposed regions may experience the out-migration of younger, more educated, or more economically dynamic individuals, while relatively more disadvantaged or politically disillusioned residents remain. If the remaining population is systematically more skeptical about whether social benefits can actually generate equality, then the estimated relationship between the China shock and welfare attitudes may be pushed upward. In this case, the OLS coefficient would partly capture compositional changes in the population of exposed regions rather than a pure attitudinal response to trade exposure.

Problem VII (c)

The OLS estimates discussed in point (b) cannot be interpreted causally because the regional China shock is potentially endogenous to French welfare attitudes. Following Colantone and Stanig (2018), there are two main concerns. First, positive local demand shocks could at the same time raise imports and shape attitudes toward redistribution and the welfare state, producing a spurious correlation. Second, the regional pattern of import competition may also reflect industrial-policy choices made in response to the very political and ideological dispositions we are trying to explain. Either way, OLS does not identify the causal effect of trade exposure on the attitudes towards the equalising role of the welfare state.

IV strategy. Following Colantone and Stanig (2018), we instrument the French regional China shock with an analogous shock built on Chinese exports to the United States, weighted by the same pre-sample French regional employment shares (`china_shock_us`). The instrument captures only the supply-side, China-driven component of import expansion that is common to France and to the United States.

The instrument must satisfy two conditions.

1. Relevance: Chinese exports to France and to the United States share the same upstream determinants: productivity catch-up, WTO accession, sectoral specialisation in low-cost manufacturing. So the two shocks should be strongly positively correlated.

2. Exogeneity: it is hard to see how American demand for Chinese goods could affect French welfare attitudes other than through the China-side supply variation already absorbed by `china_shock`; in particular, US demand is plausibly orthogonal to French local labour-market conditions and political dynamics.

Results. In the second stage, the coefficient on the instrumented China shock is -0.781 (s.e. 0.360, $p = 0.030$), against an OLS estimate of -0.796 (s.e. 0.359, $p = 0.039$). The first-stage relationship is strong, with a coefficient of 0.092 on the US-based China shock instrument and a Kleibergen–Paap F-statistic of 3457.16. The two estimates are very close in magnitude, which suggests that any simultaneity bias in OLS is small and that the negative effect of the China shock on `sbeqsoc` is robust to the instrumental-variables correction. Both coefficients are statistically significant at the 5% level. As a sensitivity check, we also report the weak-instrument-robust Anderson–Rubin Wald test, which yields $F(1, 20) = 4.32$ ($p = 0.051$), which is borderline at the 5% level and significant at the 10% level. Given that inference relies on only 21 NUTS-2 clusters, this caveat is worth bearing in mind; nonetheless, the IV estimates suggest that higher trade exposure is associated with lower values of `sbeqsoc`; given the coding of the variable, this corresponds to stronger agreement that social benefits and services lead to a more equal society.

Caveat on inference Since the analysis is restricted to French respondents, standard errors are clustered over a relatively small number of NUTS2 regions. At the same time, the specification includes several individual and regional controls, which makes the cluster-robust inference less precise. For this reason, the statistical significance of the estimates should be interpreted with some caution, although we keep the specification required by the assignment.

Table 16: VII(c): Effect of the China shock on `sbeqsoc`

Variables	(1) OLS	(2) 2SLS
Panel A: Second stage — dep. var. `sbeqsoc`		
China shock	-0.796^{**}	-0.781^{**}
	(0.359)	(0.360)
Panel B: First-stage diagnostics		
China shock IV (US-from-China)		0.092^{***}
		(0.002)
Kleibergen–Paap F		3457.16
Survey weights	YES	YES
Demographic controls	YES	YES
Education FE	YES	YES
NUTS-1 FE	YES	YES
Clustered SEs, NUTS-2	YES	YES
Observations	2,043	2,043
R^2	0.056	0.056

Notes: Standard errors clustered at the NUTS-2 level are reported in parentheses. The first-stage coefficient and Kleibergen–Paap statistic are reported as diagnostics for the 2SLS specification. The first-stage regression is estimated on the available instrument sample, but the reported OLS and 2SLS coefficients use the common second-stage sample of 2,043 observations. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Problem VII (d)

The 2SLS coefficient on china shock is negative (-0.781) and statistically significant at the 5% level. Since `sbeqsoc` is coded with 1 = agree strongly and 5 = disagree strongly with the statement that social benefits and services lead to a more equal society, a negative coefficient means that respondents living in French regions more heavily exposed to Chinese import competition are, on average, more inclined to endorse the equalising role of the welfare state. Using the standard deviation of the China shock within the regression sample, approximately 0.059, the implied one-standard-deviation IV effect is roughly -0.046 points on the 5-point scale. While modest in absolute terms — roughly 1.7% of the sample mean — the effect is consistent across OLS and IV specifications. The effect is statistically detectable under standard inference, though the weak-instrument-robust Anderson–Rubin test is borderline at the 5% level, as discussed above.

Trying to explain this finding requires delving into social scientific literature on sociopolitical attitudes, economic shocks and the complex relation between the two. Analyzing the interaction between cultural and economic phenomena is methodologically tricky, as Agnolin and colleagues highlight. The problem is the sticky and post-treatment nature of cultural variables, and the high degree of interrelation with economic ones. For this reason, we should be wary of literature drawing conclusions on the culture-economy relation by simply using cultural variables as controls (Agnolin et al. 2024). Nevertheless, the literature on the topic suggests valid theories and potential mechanisms that are of use in explaining the observed relationship.

Compensation hypothesis This first mechanism frames the increase in appreciation of the welfare state as an insurance mechanism against the distress generated by trade-triggered competitive pressures. Intuitively, as individuals interiorize the costs of the open economy, they start

asking more protection from the state, and appreciate more its role in promoting equality and redistribution. This mechanism echoes the embedded-liberalism logic by Ruggie (1982), according to which trade liberalization is politically sustainable only when paired with domestic social protection. This is also consistent with the long-standing empirical finding by Cameron (1978) and Rodrik (1998) that more open economies tend to run higher levels of public spending, precisely because exposure to external risk raises the demand for insurance against labor-market volatility. At the micro-founded level, Walter (2010) provides evidence supporting the compensation hypothesis in the context of globalization and renewed support for social protection. Under this reading, French respondents in regions more exposed to Chinese import competition place greater value on the welfare state as an insurance device.

Political scripts and translation of attitudes into preferences At first glance, a negative coefficient on China shock may seem in tension with Colantone and Stanig's central result that import exposure pushes European voters towards nationalist and radical-right parties, given that support for the equalizing role of the welfare state are usually associated with left-wing parties (Colantone and Stanig, 2018). However, we believe the findings to be compatible. The authors study vote choice, while `sbeqsoc` captures an attitudinal demand for welfare-state redistribution. The two outcomes correspond to different points along the same political-economy chain: trade-exposed individuals first develop a stronger demand for compensation, and only when mainstream parties fail to deliver on the embedded-liberalism bargain (Rodrik, 1997) does that demand convert into support for economic-nationalist platforms. What we estimate here is simply the upstream attitudinal half of that sequence.
An attempt to explain this apparent discrepancy comes from the theory on political scripts and the translation of attitudes into political preferences. Intuitively, the process is essentially threefold. Firstly, economic shocks generate structural divides creating winners and losers. Secondly, such structural changes are not deterministically translated into changes in preferences, rather, they are interpreted through the lens of culture: economic insecurity activates the historically available language related to social protection and economic rights. In the French context, this is especially plausible because the welfare state, public services, and social equality are politically familiar scripts through which economic vulnerability can be articulated, categorized, and translated into policy action (Levy, 2017). For this reason, the attitudinal reaction to economic shocks is ultimately an empirical question. Lastly, parties' construction and framing of the political agenda matters in the way changing attitudes activated by economic shocks are translated into the ultimate voting choice. Kriesi et al.'s globalization-cleavage framework is useful here: globalization creates a structural divide between winners and losers, but parties and institutions determine whether this divide is politicized as redistribution, protectionism, nationalism, or some combination of these (Kriesi et al. 2006).

Sociotropic transmission Another underlying mechanism is mobilized to explain the diffuse effects of relatively concentrated costs. Not only workers that are directly impacted by globalization modify their attitudes towards the welfare state, but also other individuals in their proximity and in society more generally. As Colantone and Stanig (2018) emphasise, the political response to the China shock is largely sociotropic: voters react to community-level economic distress observed in their region rather than only to their personal labour-market situation. Respondents living in regions specialised in industries vulnerable to Chinese competition can therefore update their views on the equalising role of the welfare state simply by observing the local consequences of trade exposure, even when they are themselves employed in non-tradable sectors.

Problem VII (e)

In the final point of Problem VII we augment the ESS-based dataset constructed in points (a)–(d) with information from the Comparative Manifesto Project (CMP), in order to associate each respondent with a measure of the corporatist orientation of the party they voted for in the last national election. The relevant CMP variable is `per405`, defined in the codebook as "favourable mentions of cooperation of government, employers, and trade unions simultaneously". The variable captures the salience of corporatist, mixed-economy ideas in a party's manifesto.

Procedure. We restrict the CMP file (`MPDataset_MPDS2025a.dta`) to French parties competing in the 2012 legislative election (`countryname == "France"` and `date == 201206`) and we keep the variables `party`, `partyname`, `partyabbrev`, and `per405`. The merge with the ESS sample uses the variable `prtvtcfr`, which records the party voted for in the last French national election. Since textual party labels do not coincide across the two datasets, the matching is performed manually on the basis of party identity rather than name strings: we generate the CMP party identifier `party_cmp` within the ESS dataset following the mapping reported in Table 17, and then merge ESS to CMP on `party_cmp`.

Table 17: VII(e): Matching ESS `prtvtcfr` to CMP `party_cmp` for France, 2012

ESS code	CMP id	Party	Abbrev.	per405
1	31630	Nouveau Centre	NC	0.00
2	31720	Front National	FN	0.00
3	31421	Parti Radical Valoisien	PR	0.00
6	31021	Front de Gauche	FDG	0.00
7	31230	Parti Radical de Gauche	PRG	0.00
9	31320	Parti Socialiste	PS	0.00
10	31626	Union pour un Mouvement Populaire	UMP	0.21
11	31624	Mouvement Démocrate	MoDem	0.48
12	31110	Europe Écologie Les Verts	EELV	0.53

Source: own elaboration based on ESS Round 8 and Comparative Manifesto Project (MPDS2024a).
`per405`: "Favourable mentions of cooperation of government, employers, and trade unions simultaneously."

After the merge, exactly nine distinct CMP parties are matched to ESS respondents, in line with the assignment's expectation. The Centrist Alliance (AC) is present in the CMP 2012 French wave but does not appear among the parties listed in the ESS variable `prtvtcfr`, so it is dropped from the merged sample. Other ESS response codes (e.g. "other", "did not vote") are not associated with any CMP party and are similarly excluded. The resulting augmented dataset, saved as `problem7_e.dta`, contains both the regional China shock built in Problems VI–VII and the party-level corporatism score, and constitutes the natural input for any subsequent analysis linking trade exposure to the corporatist content of voting choices.

Reading the per405 distribution. Six of the nine matched parties report a `per405` score equal to zero, meaning that their 2012 manifesto contains no quasi-sentence explicitly favourable to tripartite cooperation between government, employers, and trade unions. The remaining three—UMP (0.21), MoDem (0.48), and EELV (0.53)—span the centre-right, centre, and green political families, and devote a small but positive share of their manifestos to corporatist themes. Notably, neither the radical left (FDG) nor the radical right (FN) places any emphasis on corporatist cooperation. The variable should therefore be read as a focused measure of corporatist rhetoric rather than as a generic left–right indicator: parties that converge on a corporatist discourse can otherwise differ substantially on redistribution, public ownership, or trade policy.

Problem VII (f)

We next examine whether regional exposure to the China shock is associated with voting for parties with a more corporatist or mixed-economy orientation. Following the assignment, we attach to each respondent's party voted in the last national election the Manifesto Project score measuring favourable mentions of cooperation between government, employers, and trade unions. The dependent variable is transformed as:

$$\text{corporatism score}_{pc} = \log(0.5 + z_{pc}),$$

where z_{pc} is the Manifesto Project score of the voted party.

Table 18 reports the results. In the OLS specification, the coefficient on the China shock is positive and statistically significant ($\hat{\beta} = 0.396$, $p < 0.01$). In the IV specification, where the

regional China shock is instrumented with the US-based import shock, the coefficient remains positive and highly significant ($\hat{\beta} = 0.404$, $p < 0.01$). The two estimates are very similar in magnitude, suggesting that the positive relationship is robust to using the instrumental-variable strategy.

The first stage is very strong. The coefficient on the instrument `china_shock_us` is positive and precisely estimated, and the Kleibergen–Paap F-statistic is equal to 3337.38, well above conventional weak-instrument thresholds. This supports the relevance of the instrument, although the IV interpretation still relies on the validity of the exclusion restriction.

Overall, the evidence indicates that respondents living in regions more exposed to Chinese import competition are more likely to vote for parties whose manifestos place greater emphasis on corporatist coordination and the mixed economy. In substantive terms, stronger trade exposure appears to be associated with support for political platforms favouring cooperation between the state, employers, and labour, rather than purely market-based adjustment.

Table 18: China shock and corporatism score of voted party

	(1)	(2)	(3)
Variables	**OLS**	**2SLS**	**First stage**
China shock	0.396***	0.404***	
	(0.114)	(0.111)	
China shock instrument (US-from-China)			0.092***
			(0.002)
Observations	957	957	957
R-squared	0.090	0.090	0.998
Kleibergen-Paap F		3337.381	

Robust standard errors clustered at NUTS2 level in parentheses.

All regressions include gender, age, education dummies, NUTS1 dummies, and survey weights.

$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

These results should nevertheless be interpreted with caution. The analysis is restricted to matched party voters in France, the number of regional clusters is limited relative to the number of controls, and the corporatism score is equal to zero for many parties. Therefore, the estimates should be read as suggestive evidence on the relationship between trade exposure and the ideological profile of the voted party, rather than as definitive causal evidence.[2]

Problem VII (g)

Taken together, the results suggest that the political reaction to globalization is not limited to a simple anti-trade or anti-immigration backlash. In our French sample, stronger regional exposure to Chinese import competition is positively associated with voting for parties whose platforms place more emphasis on corporatist coordination and the mixed economy. The results on attitudes toward the equalising role of social benefits point in a broadly similar direction: trade-exposed respondents are more likely to agree that social benefits and services reduce inequality. The IV estimate is statistically significant under standard inference, although the weak-instrument-robust Anderson–Rubin test is borderline at the 5% level, so the causal interpretation should still be treated with caution.

This pattern is consistent with the compensation hypothesis. When globalization generates visible local labour-market risks, voters may demand stronger forms of insurance, redistribution, and institutional coordination. Rather than reacting only through nationalist or anti-system channels, respondents in trade-exposed regions may also support political platforms that promise a more managed form of economic adjustment, based on cooperation between the state, employers, and labour.

[2] As a robustness check, we also re-estimated the regressions after grouping education into three broader categories. The coefficient on the China shock remains positive, statistically significant, and very similar in magnitude: it decreases only from 0.396 to 0.375 in OLS, and from 0.404 to 0.386 in IV. This suggests that the main result is not driven by the highly disaggregated education controls.

In this sense, the globalization backlash should not be interpreted narrowly as support for protectionism alone. It may also take the form of increased demand for domestic compensation, social protection, and coordinated economic governance—what Ruggie (1982) describes as embedded liberalism. Our results therefore complement the standard backlash narrative by showing that trade shocks can be associated not only with political discontent, but also with support for parties advocating compensation and corporatist mediation.

Appendix

Table 19: Mean log LP TFP by country and sector

Group	**Mean**	**Median**	**Std. Dev.**	**N**
France, Sector 22	3.491	3.482	0.304	796
France, Sector 36	2.565	2.590	0.363	85
Italy, Sector 22	3.572	3.531	0.508	539
Italy, Sector 36	2.509	2.491	0.361	1,116
Total	3.045	3.066	0.633	2,536

Log LP TFP from the cleaned sample (Problem IV(a)).

References

1. Agnolin, P., Colantone, I., & Stanig, P. (2024). In search of the causes of the globalization backlash: Methodological considerations on post-treatment bias. *FEEM Working Paper No. 28.2024*. FEEM Working Paper
2. Cameron, D. R. (1978). The expansion of the public economy: A comparative analysis. *American Political Science Review*, 72(4), 1243–1261. doi:10.2307/1954537
3. Cobb, C. W., & Douglas, P. H. (1928). A theory of production. *American Economic Review*, 18(1), 139–165.
4. Colantone, I., & Stanig, P. (2018). The trade origins of economic nationalism: Import competition and voting behavior in Western Europe. *American Journal of Political Science*, 62(4), 936–953. doi:10.1111/ajps.12358
5. De Loecker, J., & Warzynski, F. (2012). Markups and firm-level export status. *American Economic Review*, 102(6), 2437–2471. doi:10.1257/aer.102.6.2437
6. European Social Survey European Research Infrastructure. (2023). *ESS8: European Social Survey Round 8 data, edition 2.3*. Sikt – Norwegian Agency for Shared Services in Education and Research. doi:10.21338/ess8e02_3
7. Kriesi, H., Grande, E., Lachat, R., Dolezal, M., Bornschier, S., & Frey, T. (2006). Globalization and the transformation of the national political space: Six European countries compared. *European Journal of Political Research*, 45(6), 921–956. doi:10.1111/j.1475-6765.2006.00644.x
8. Lehmann, P., Franzmann, S., Al-Gaddooa, D., Burst, T., Ivanusch, C., Regel, S., Riethmüller, F., Volkens, A., Weßels, B., & Zehnter, L. (2024). *The Manifesto Data Collection. Manifesto Project (MRG/CMP/MARPOR), version 2024a*. Wissenschaftszentrum Berlin für Sozialforschung. doi:10.25522/manifesto.mpds.2024a
9. Lehmann, P., Franzmann, S., Al-Gaddooa, D., Burst, T., Ivanusch, C., Regel, S., Riethmüller, F., Volkens, A., Weßels, B., & Zehnter, L. (2024). *Manifesto Project Dataset codebook, version 2024a*. Wissenschaftszentrum Berlin für Sozialforschung.
10. Levinsohn, J., & Petrin, A. (2003). Estimating production functions using inputs to control for unobservables. *Review of Economic Studies*, 70(2), 317–341. doi:10.1111/1467-937X.00246
11. Levy, J. D. (2017). The return of the state? France's response to the financial and economic crisis. *Comparative European Politics*, 15(4), 604–627. doi:10.1057/cep.2015.36
12. Melitz, M. J. (2003). The impact of trade on intra-industry reallocations and aggregate industry productivity. *Econometrica*, 71(6), 1695–1725. doi:10.1111/1468-0262.00467
13. Rodrik, D. (1997). *Has globalization gone too far?* Washington, DC: Institute for International Economics.
14. Rodrik, D. (1998). Why do more open economies have bigger governments? *Journal of Political Economy*, 106(5), 997–1032. doi:10.1086/250038
15. Ruggie, J. G. (1982). International regimes, transactions, and change: Embedded liberalism in the postwar economic order. *International Organization*, 36(2), 379–415. doi:10.1017/S0020818300018993
16. Walter, S. (2010). Globalization and the welfare state: Testing the microfoundations of the compensation hypothesis. *International Studies Quarterly*, 54(2), 403–426. doi:10.1111/j.1468-2478.2010.00593.x
17. Wooldridge, J. M. (2009). On estimating firm-level production functions using proxy variables to control for unobservables. *Economics Letters*, 104(3), 112–114. doi:10.1016/j.econlet.2009.04.026